

SYDNEY BANKSTOWN, Australia

WMO: 947650

Lat: 33.9181S

Lon: 150.9864E

Elev: 8

StdP: 101.23

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.3	4.4	-3.3	2.9	17.0	-1.5	3.3	15.0	10.3	15.4	9.4	16.2	1.0	340	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.2	34.4	20.4	32.0	20.5	30.0	20.1	23.3	29.5	22.5	28.0	21.8	26.8	5.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.6	16.2	25.4	20.9	15.6	24.6	20.2	14.9	24.1	69.4	29.6	66.3	28.0	63.8	26.9	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.5	7.6	0.8	40.4	0.9	2.9	0.1	42.4	-0.4	44.1	-0.9	45.7	-1.6	47.8		
(5)			0.2	24.7	0.9	0.9	-0.4	25.3	-0.9	25.8	-1.4	26.3	-2.1	27.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	23.2	22.8	21.2	18.1	14.9	12.4	11.4	12.6	15.3	17.8	19.9	21.8		
	DBStd	4.78	2.91	2.67	2.40	2.33	2.14	1.89	1.75	2.17	2.63	2.99	3.25	3.04		
	HDD10.0	9	0	0	0	0	0	2	5	2	0	0	0	0		
	HDD18.3	880	1	1	3	31	108	178	213	178	97	47	19	4		
	CDD10.0	2779	408	358	347	242	152	74	50	83	160	242	296	366		
	CDD18.3	608	151	125	92	23	2	0	0	1	7	31	64	113		
	CDH23.3	4378	1103	824	484	150	16	0	1	14	113	329	516	826		
	CDH26.7	1504	421	267	121	31	1	0	0	2	29	114	204	315		
(14) Wind	WSAvg	3.1	3.7	3.5	3.1	2.5	2.5	2.5	2.6	3.0	3.2	3.4	3.7	3.7		
(15) Precipitation	PrecAvg	911	94	107	118	94	67	81	45	37	43	64	81	69		
	PrecMax	1398	233	440	334	416	172	325	150	194	162	190	213	261		
	PrecMin	493	4	16	13	4	3	7	0	2	1	1	12	8		
	PrecStd	269	68	99	84	102	50	76	43	49	39	61	52	58		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.3	36.7	32.7	30.2	25.3	21.1	22.3	25.2	30.1	33.3	36.0	36.7		
		MCWB	21.5	21.4	20.8	18.2	16.1	14.2	13.2	14.2	15.9	16.9	18.5	20.6		
	2%	DB	33.6	32.2	29.5	26.7	23.0	19.2	19.3	22.1	26.5	29.8	31.9	32.9		
		MCWB	21.6	22.0	20.5	17.9	15.7	13.5	11.9	13.0	14.9	16.8	19.5	20.3		
	5%	DB	31.0	29.6	27.7	24.8	21.3	18.1	17.8	20.1	23.9	27.0	28.5	30.1		
		MCWB	21.5	21.4	20.1	17.6	14.9	12.9	11.6	12.5	14.5	16.6	18.8	19.8		
	10%	DB	28.7	27.7	25.9	23.2	19.9	17.0	16.5	18.4	21.7	24.3	25.9	27.6		
		MCWB	20.8	20.8	19.5	17.2	14.5	12.3	11.0	11.7	13.9	16.1	18.2	19.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.4	24.5	23.2	20.7	18.6	16.3	15.5	16.2	18.0	20.1	22.1	23.4		
		MCDB	32.3	31.1	28.0	25.5	21.1	18.6	18.8	21.6	24.0	26.0	29.6	30.4		
	2%	WB	23.2	23.4	21.9	19.6	17.6	15.1	13.7	14.7	17.0	18.9	21.0	22.1		
		MCDB	30.1	28.8	26.4	24.1	20.1	17.4	16.5	19.3	22.4	24.9	27.2	28.6		
	5%	WB	22.4	22.5	21.2	18.8	16.5	14.2	12.8	13.7	16.0	18.0	20.1	21.3		
		MCDB	28.3	27.2	25.3	22.6	19.4	16.6	15.8	18.1	21.2	23.4	25.7	27.1		
	10%	WB	21.7	21.7	20.5	18.1	15.5	13.4	11.9	12.7	15.1	17.2	19.1	20.5		
		MCDB	26.7	26.0	24.4	21.8	18.8	15.9	15.2	16.9	20.0	22.3	24.2	25.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	8.7	8.9	10.0	10.5	9.8	11.1	12.0	11.9	11.1	9.9	9.6		
		MCDBR	14.4	12.6	12.3	13.0	12.7	11.5	13.5	15.1	16.7	16.9	15.7	14.8		
	5% WB	MCWBR	4.6	4.2	4.6	5.5	6.1	6.4	7.3	7.3	6.8	6.1	5.3	4.8		
		MCWBR	11.5	10.2	9.9	10.6	9.5	8.7	10.4	12.7	12.9	12.4	12.3	12.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.398	0.396	0.363	0.336	0.319	0.311	0.301	0.309	0.343	0.362	0.387	0.399		
		taud	2.442	2.467	2.541	2.593	2.587	2.608	2.617	2.586	2.492	2.467	2.438	2.416		
	Ebn at Noon	Ebn at Noon	940	921	916	889	850	833	864	905	922	943	944	943		
		Edh at Noon	121	114	99	83	74	68	71	82	102	112	121	125		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.47	5.54	4.73	3.82	3.01	2.38	2.79	3.69	4.91	5.70	6.08	6.63		
	RadStd	RadStd	0.48	0.49	0.41	0.33	0.20	0.22	0.22	0.21	0.28	0.50	0.55	0.56		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.52	N/A	N/A	N/A	N/A	N/A	-5	-102	+211	+112
(47) Regional (34 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	-2	-85	+184	+98

Nomenclature: See separate page